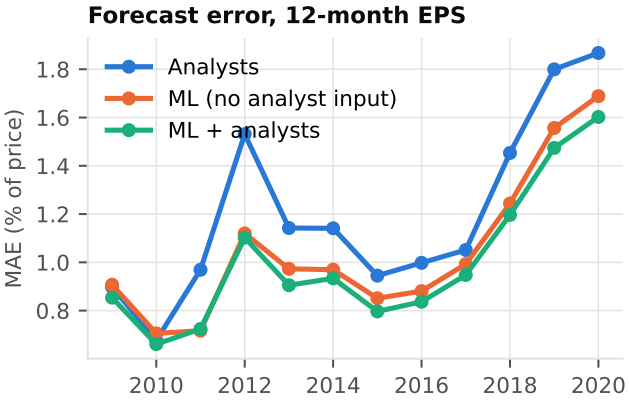
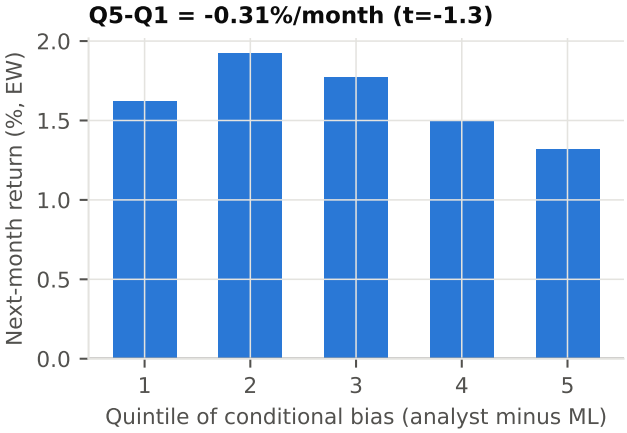
Idea 5 PoC — A machine benchmark for Chinese analysts

Proof of concept, October 2026. Code: `idea05_ml_analyst_bias/run.py`.

Question and design

Are Chinese consensus forecasts biased *conditionally*, relative to a look-ahead-free machine forecast? Does that bias predict returns, and with the negative US sign (van Binsbergen et al., 2023) or the positive sign reported for China (Liu et al., 2023)? The forecast is the I/B/E/S 12-month-forward consensus scaled by price, F_m. The outcome is the I/B/E/S trailing EPS twelve months later scaled by the same price, R_m. Keeping the outcome inside I/B/E/S avoids the bonus-share basis mismatch with the quarterly EPS. An XGBoost model predicts R_m from public data only: trailing EPS/P, B/M, size, past returns, turnover, the latest surprises and streak, and TTM EPS growth. It is retrained every January, using only rows whose outcome had already been realised. Conditional bias is $F_m - \widehat{R}_m^{ML}$. Test sample: 181,482 covered firm-months, 2,860 firms, 2009–2020.





	Analysts	ML, no analyst input	ML + analysts
MAE (% of price)	1.19	1.02	0.98
Mean error $R - f$ (% of price)	−0.66	−0.05	−0.06
Share of firm-months beating analysts	–	54.9%	58.1%

Bias quintile portfolios (next month)	Q5−Q1 raw	CH-3 α	CH-4 α	FM slope per 1 SD of bias
Equal-weighted	−0.31 (−1.3)	−0.71 (−2.6)	−0.83 (−3.0)	−0.15 (−3.0)
Value-weighted	−0.18 (−0.5)	−0.41 (−1.1)	−0.59 (−1.7)	

% per month (t-statistics, Newey–West). The FM regression controls for size, B/M, momentum, short-term reversal and turnover.

Findings

- **A simple machine beats the consensus.** Without seeing any analyst input, XGBoost lowers the MAE by 14% and is more accurate in every year from 2011. Adding the consensus as an input helps further, so analysts do carry information the model lacks ("man + machine").
- **Analysts are optimistic, and predictably so.** The consensus exceeds the realised outcome by 0.66% of price on average, while the machine is almost unbiased. The conditional bias predicts the analyst error almost one for one (slope −0.87, $t = -12.7$).
- **Optimism relative to the machine predicts *lower* returns, as in the US.** Stocks with the most optimistic analysts underperform by 0.83% per month on a CH-4 basis (EW). The FM slope is negative after standard controls. This is the opposite of the positive sign Liu et al. (2023) report with a statistical benchmark. The effect is weaker value-weighted and after 2015.

Caveats and next steps

(i) The panel holds only the time-weighted 12-month consensus, and the trailing "outcome" partly blends in later forecasts. Fiscal-year I/B/E/S detail (FY1 and FY2 consensus with the number of analysts) would give a clean target. (ii) The raw spread is insignificant; the alpha comes from factor loadings, so the risk adjustment needs scrutiny. (iii) Next: reconcile with Liu et al. by replicating their growth-forecast signal on the same sample, and check the look-ahead discipline that Zhang et al. (2025) show can reverse US conclusions.

References. van Binsbergen, Han and Lopez-Lira (2023, RFS); Zhang, Zhu and Linnainmaa (2025, RFS); Liu, Zhu, Zhang and Zhang (2023, PBFJ); Campbell, Ham, Lu and Wood (Management Science, forthcoming).